

Algebra 2

Unit 6: Solving Cube Root Equations

Day 1: Intro to Radical Equations

Name: _____

I can solve basic cube root and rational exponent equations.

Key Concept: Reciprocal Exponents

To solve $x^{a/b} = c$, raise both sides to the power of $\frac{b}{a}$.

- Remember: $x^{1/3}$ is the same as $\sqrt[3]{x}$.
- Remember: $x^{2/3}$ is the same as $(\sqrt[3]{x})^2$.

Part 1: Basic Isolation

1. $\sqrt[3]{x} = 5$

4. $\sqrt[3]{x+1} = 4$

2. $2\sqrt[3]{x-5} = -4$

5. $\frac{\sqrt[3]{x}}{2} = 3$

3. $\sqrt[3]{x} + 8 = 10$

6. $\sqrt[3]{x} - 7 = -10$

Part 2: Rational Exponents

1. $x^{\frac{1}{3}} = 4$

2. $x^{\frac{2}{3}} = 25$

Part 3: One-Step Operations

1. $\sqrt[3]{x+7} = 3$

3. $(x+1)^{\frac{2}{3}} = 4$

2. $\sqrt[3]{2x} = 4$

4. $(x-4)^{\frac{2}{3}} = 16$

You Try!

1. $\sqrt[3]{x} = -2$

6. $(x+5)^{\frac{2}{3}} = 1$

2. $x^{\frac{1}{3}} + 5 = 8$

7. $\sqrt[3]{x+2} = 5$

3. $3\sqrt[3]{x} = 15$

8. $2\sqrt[3]{x} = 12$

4. $x^{\frac{2}{3}} = 49$

9. $(x-2)^{\frac{1}{3}} = -4$

5. $\sqrt[3]{x-10} = -3$

10. $\sqrt[3]{3x} = 3$

Answer Bank

−62
−17
−8
−6
−4
−3
7
8
9
20
27
32
63
64
68
123
125
216
343